# REOPENING CITIES, STATES AND SCHOOLS SAFELY
# VIA FREQUENT RAPID ANTIGEN TESTING

**MICHAEL MINA**, MD, PhD
mmina@hsph.Harvard.edu

@michaelmina_lab

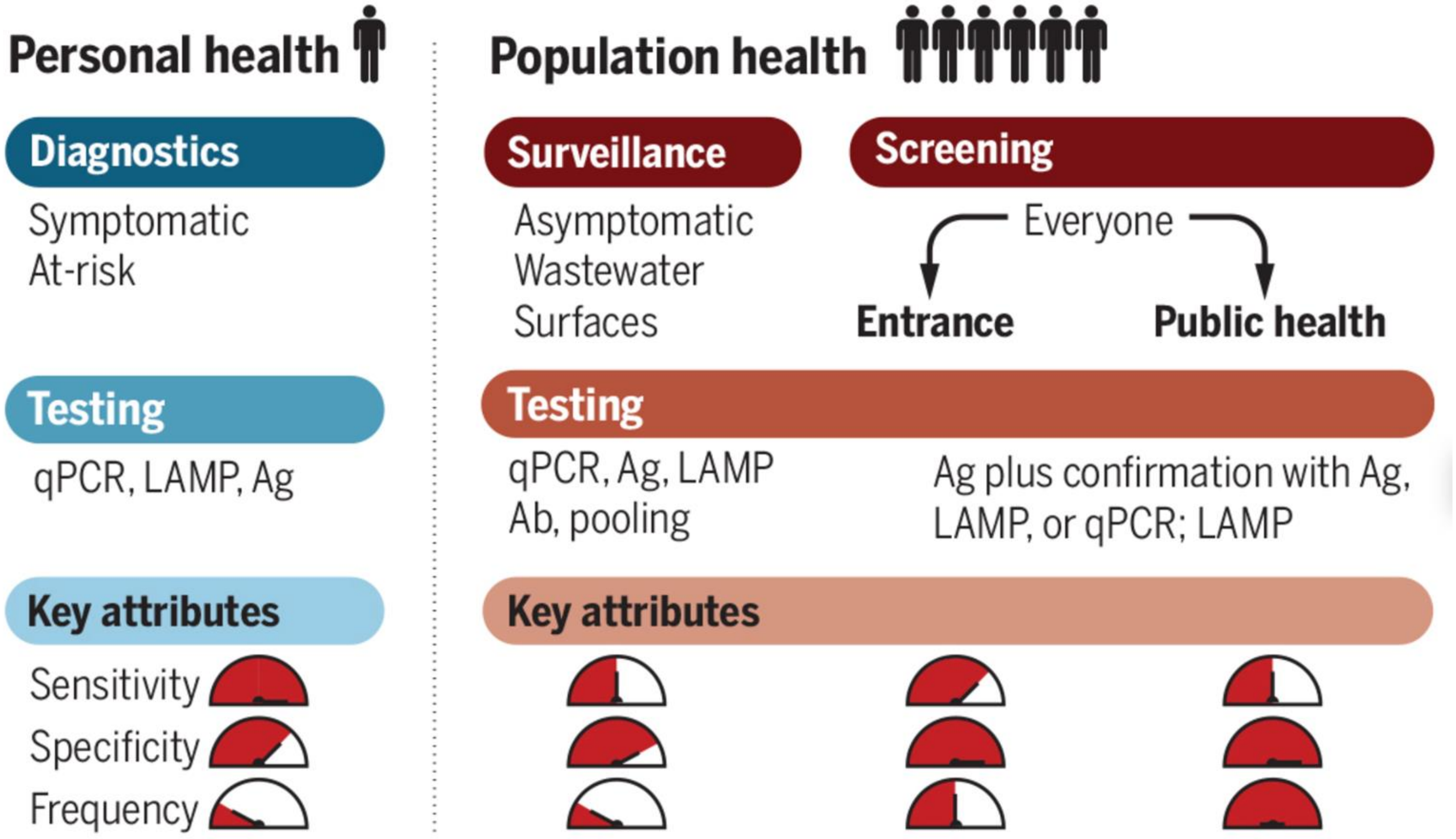
Center for Communicable Disease Dynamics

Harvard T.H. Chan School of Public Health

Harvard Medical School

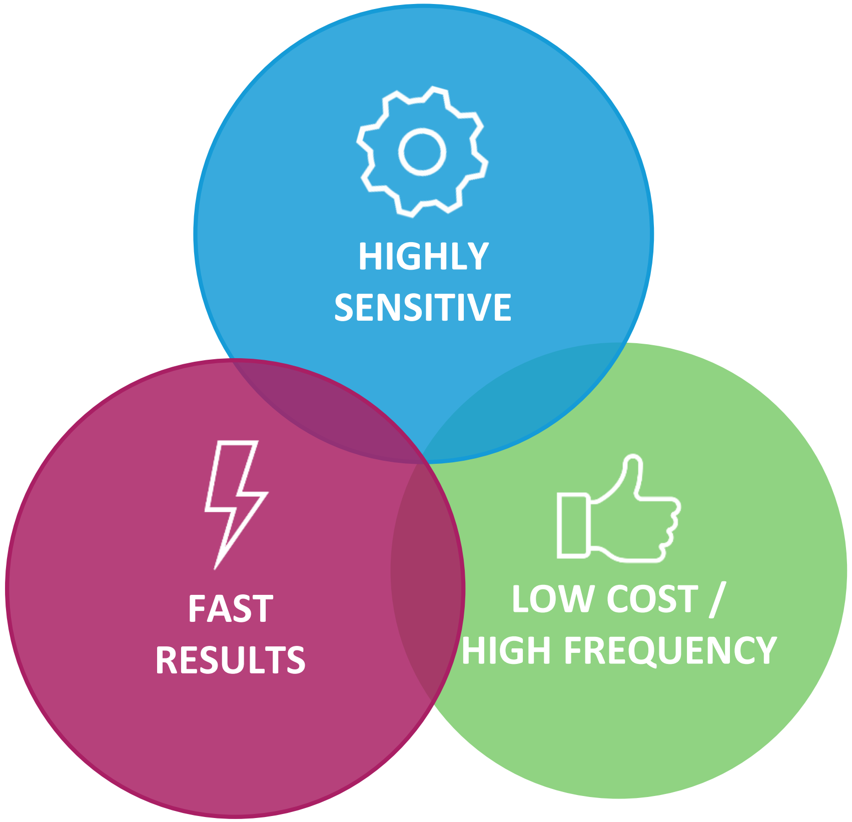
Brigham & Women's Hospital

# Testing: One Size Doesn't Fit All



# POPULATION SCREENING TEST CHARACTERISTICS FOR OUTBREAK CONTROL:

## PICK ONE OR TWO, BUT NOT THREE




USUAL PRIORITY ORDER (i.e. at FDA)

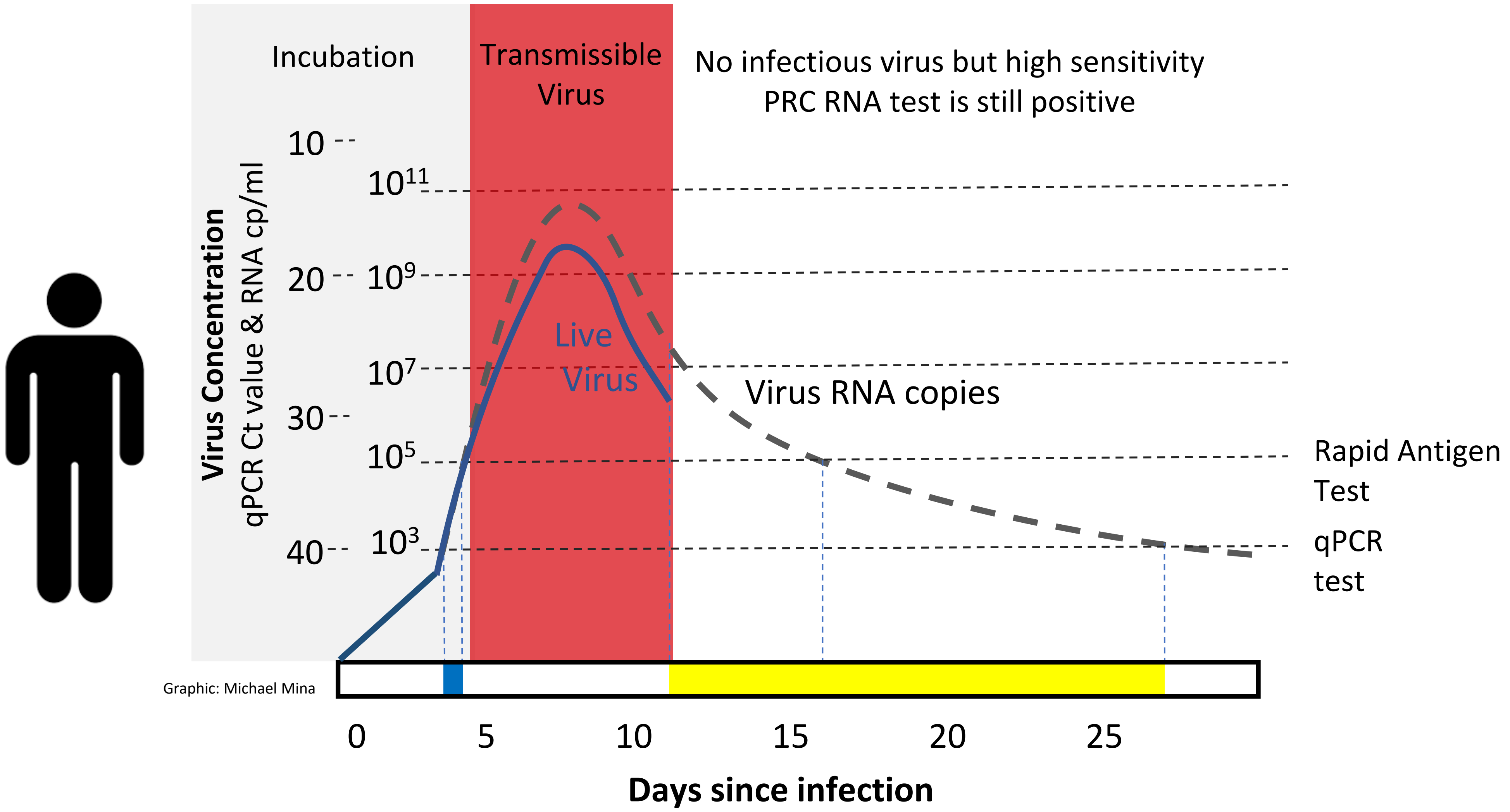
1. **Extremely high sensitivity**
2. Cost (frequency)
3. Speed to get results

But is this ***correct***?

- What about **<u>Fast</u> & <u>in</u>expensive (use daily), but… 1000x less sensitive?**
- *(Important because such tests can be available today…)*

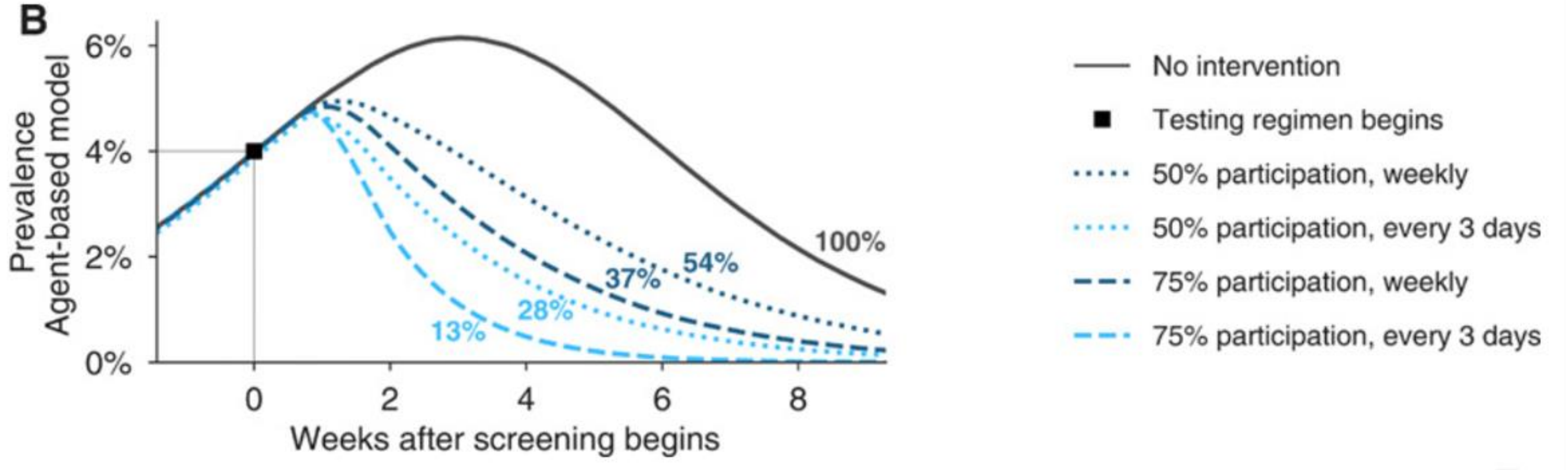
# VIRUS KINETICS WITHIN THE INDIVIDUAL

## TO ANSWER THE QUESTION…LOOK TO THE VIRUS KINETICS




Graphic: Michael Mina

# Frequent Rapid Tests in a fraction of the population can reduce incidence quickly




Right now, 100 people become 500 new infections in 4 weeks, at R = 1.4

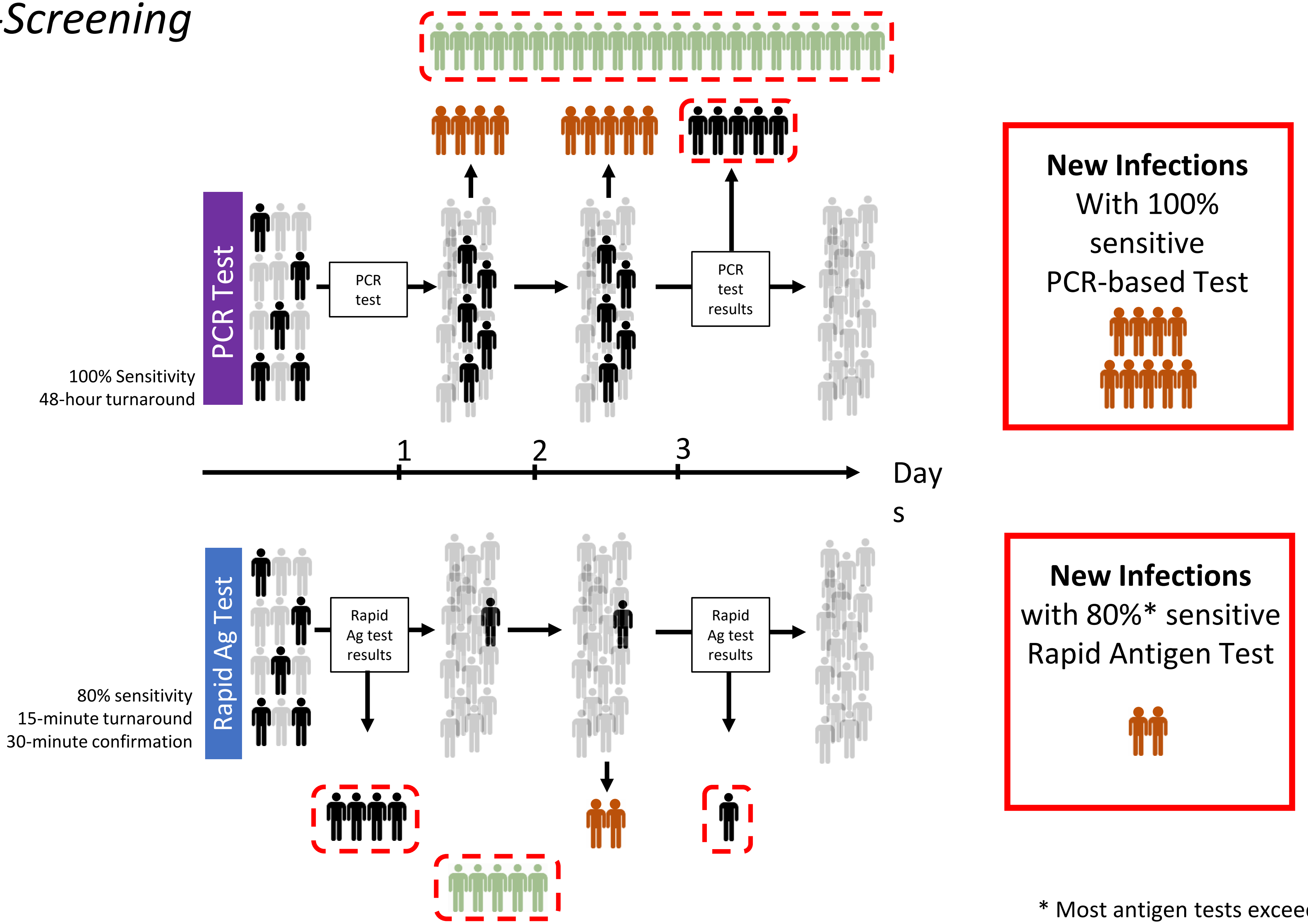
**If we get 100 people to just infect 90**, 100 people become 20 in 4 weeks.

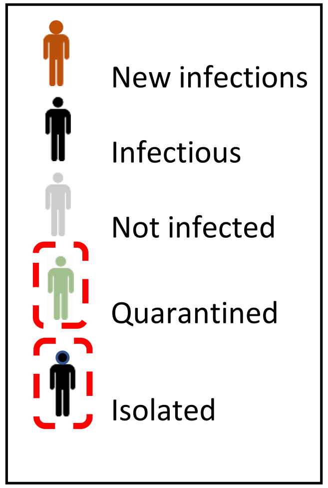
**96% Improvement in 4 weeks.**

Larremore… Mina, Science Advances, 2020

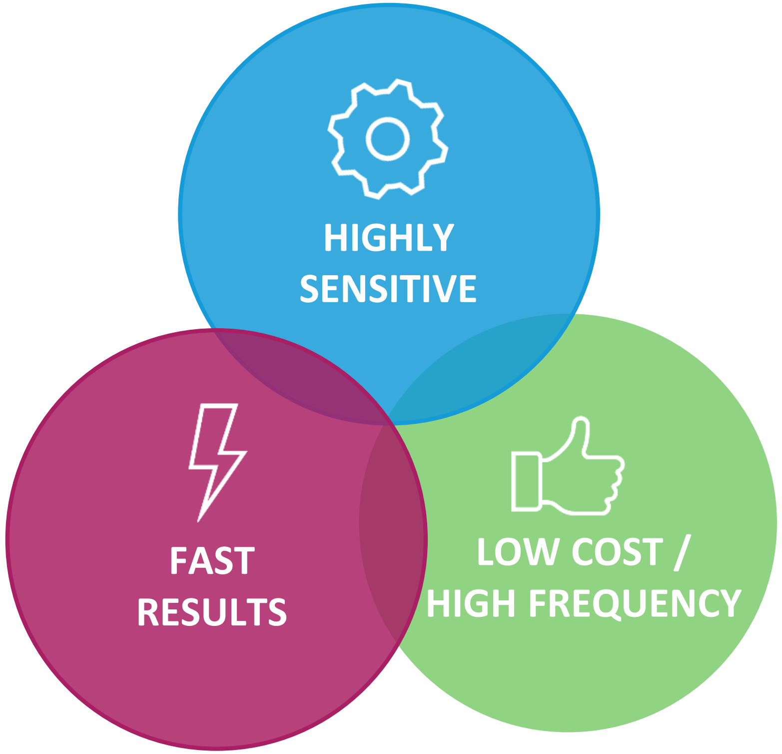
# Example of School Based Testing

*Entrance-Screening*




* Most antigen tests exceed 80% and approach 95% for high viral loads





The New York Times

Opinion

# A Cheap, Simple Way to Control the Coronavirus

With easy-to-use tests, everyone can check themselves every day.

By Laurence J. Kotlikoff and Michael Mina

Mr. Kotlikoff is a professor of economics at Boston University and Dr. Mina is an assistant professor of epidemiology at the Harvard T.H. Chan School of Public Health.

## USUAL PRIORITY ORDER

1. **Sensitivity**
2. Cost (frequency)
3. Speed to get results

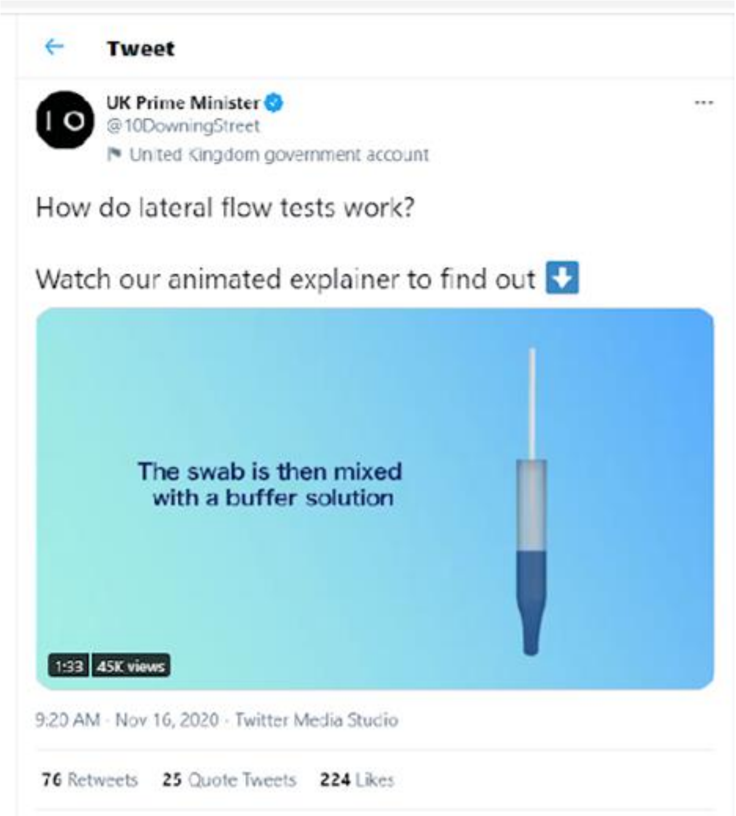
## OPTIMAL PRIORITY ORDER

1. **Cost (frequency)**
2. **Speed to get results**
3. Sensitivity

In order to rapidly detect and prevent outbreaks:

- It is OK to **sacrifice the high sensitivity of a molecular test** in exchange for higher frequency and faster results

# Rapid antigen testing is an inexpensive, equitable solution for public health departments. Follow the UK's example…




This video was created by the UK government to educate British citizens prior to the launch of their massive public health screening program this past fall.

# Recommended Applications of Rapid Testing

| Rapid Testing Application | Location | Frequency | Implementation Support | Implementation Challenges |
|---|---|---|---|---|
| Population Screening<br><br>**To reduce overall prevalence at the community level** | • At-home<br>• Surveillance studies | • 2x/week for at least 50% of population for first 12 weeks<br>• Portion of population continues testing for quick detection of new outbreaks | • Should be promoted through public education and communication efforts | • Voluntary uptake<br>• Public health reporting<br>• Acting on results<br>• Price to user<br>• Financial implications of isolation |
| Entrance Screening*<br><br>**To improve daily safety within a particular location** | • Schools<br>• Workplaces<br>• Retail stores<br>• Sports & entertainment venues<br>• Airports & transportation<br>• Borders & ports of entry<br>• Prisons<br>• Long-term care | • 3x/week for all teachers and staff in schools for first 12 weeks<br>• 3x/week for all in-person workplaces<br>• Required for all staff and guests upon entry to venues listed | • Enabled by federal and state regulatory authority and voluntary local private and public sector agreements | • Voluntary compliance<br>• Operations/logistics |



*Enabled by federal and state regulatory authority and voluntary local private and public sector agreements

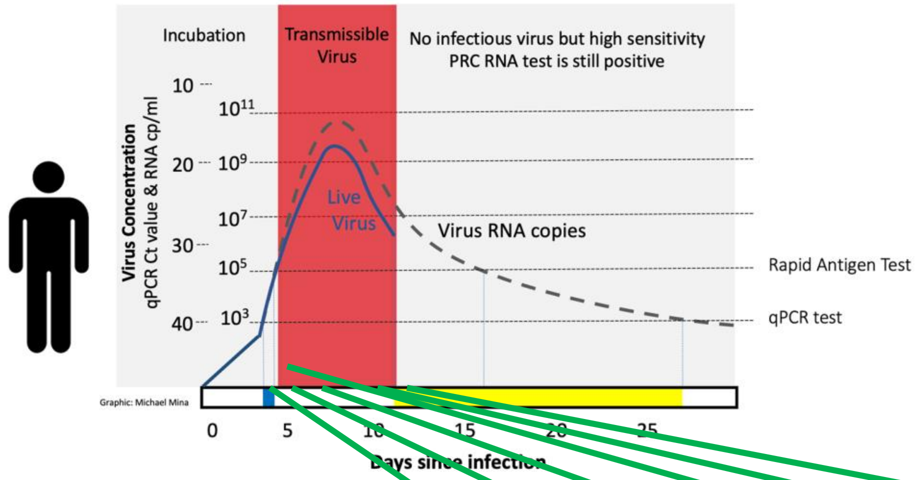
# Current Snapshot of Rapid Test Manufacturers and Capacity (With No Government Intervention)

| Test Type | Test | Price | Jan-21 | Feb-21 | Mar-21 | Apr-21 | May-21 | Jun-21 |
|---|---|---|---|---|---|---|---|---|
| *Antigen Self-Test* | **Ellume** (EUA Dec 2020) | *$30* | *0.5* | *1* | *3* | *4* | *5* | *7.5* |
| *Antigen Self-Test* | **Abbott BINAX At Home** (EUA Dec 2020) | *$25* | *6* | *10* | *10* | *30* | *30* | *30* |
| *Antigen Self-Test* | **Gauss/Cellex** (EUA submitted Q4 2020) | *$30 TBD* | *1* | *5* | *25* | *30* | *30* | *35* |
| *Antigen Self-Test* | **E25 Bio** (EUA submitted Q4 2020) | | *2* | *15* | *30* | *45* | *50* | *50* |
| *Antigen Self-Test* | **OraSure** (EUA to be submitted Q1 2021) | | *0* | *1* | *4* | *5* | *5* | *5* |
| *Antigen Self-Test* | **Quidel QuickVue at Home**<br>(Addt'l EUA for Home in 2021?) | | *0* | *4* | *4* | *4* | *4* | *4* |
| *Antigen Self-Test* | **Innova** (EUA submitted November 2020, (Addt'l EUA for Home submitting Feb 2021) | *$5* | *100* | *300* | *450* | *600*<br>*May require DPA for GE Health for NC Membrane* | *900*<br>*May require DPA for GE Health for NC Membrane* | *1200*<br>*May require DPA for GE Health for NC Membrane* |
| *Antigen Self-Test* | **Access Bio** (EUA for At-Home to be submitted Q1 2021) | *$15-$25 TBD* | *NA* | *10* | *15* | *15* | *30+* | *30+* |
| *Antigen Self-Test* | *New Tests – Other Manufacturers (e.g., Siemens / Roche / Celltrion)* | | *3* | *8* | *20* | *30* | *50* | *60* |
| **Total EUA Today & Future** | | | ***112.5*** | ***354*** | ***561*** | ***763*** | ***1,104*** | ***1,421.5*** |

# Current Rapid Antigen Testing Pilot Programs

- Liverpool, UK (and soon to be the entire UK)
- New England colleges – Colby and Harvard
- San Francisco public transit study

# Rapid antigen tests are highly sensitive to detect infectious people




College Campuses

**4200 Innova Rapid Tests Performed**

0% False Positives (100% Specificity)

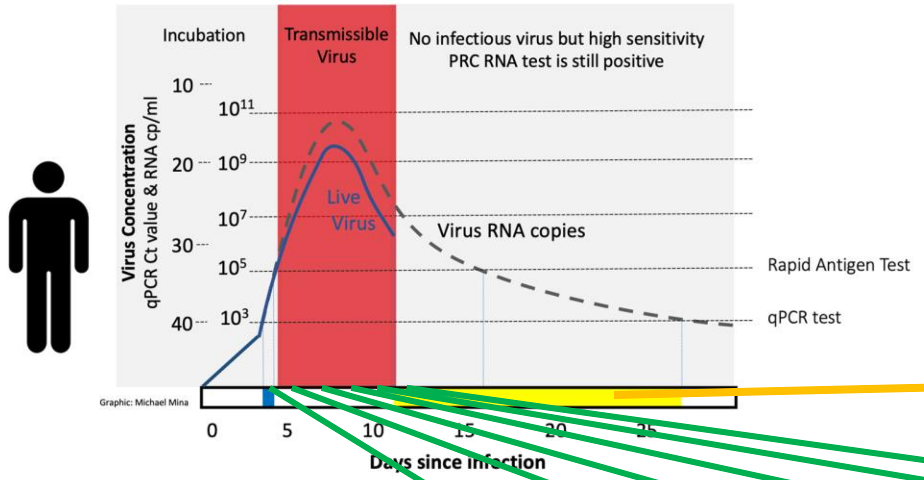
**85% Sensitivity within 24 hours of viral load exceeding PCR threshold**
**100% Sensitivity within 48 hours of viral load exceeding PCR threshold**

**Sensitivity To Detect Infected People**

| Ag Test Detected? | 0 days | 1 days | 2 days | 3 days | 4 days | 5 days |
|---|---|---|---|---|---|---|
| Detected | 7 | 17 | 21 | 22 | 22 | 24 |
| Not Yet Detected | 13 | 3 | 0 | 0 | 0 | 0 |
| Sensitivity to detect infected ppl | 37% | 85% | 100% | 100% | 100% | 100% |

*Note
Increase in numbers of cases detected was due to tracking down PCR positive individuals who did not immediately have a rapid antigen test on any of the preceding day.

# Papers that show low sensitivity include recovered, non-infectious individuals



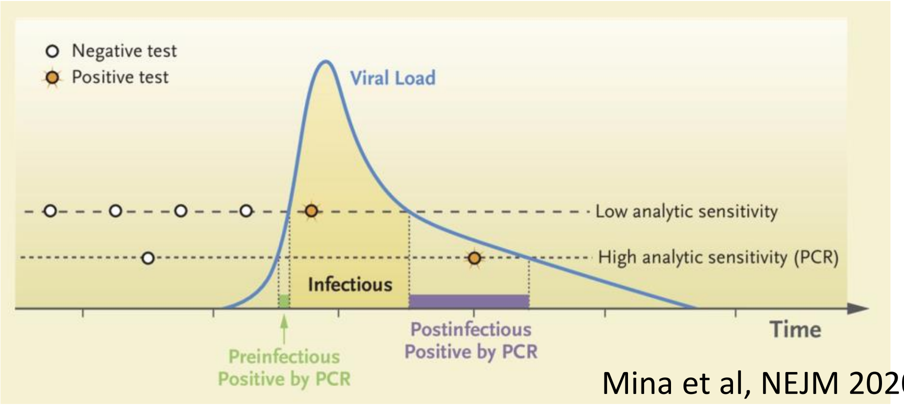

College Campuses
4200 Innova Rapid Tests Performed
0% False Positives (100% Specificity)

85% Sensitivity within 24 hours of viral load exceeding PCR threshold
100% Sensitivity within 48 hours of viral load exceeding PCR threshold
**If we add those who are already recovered from isolation (shown in the yellow region; n = 22; not shown), sensitivity drops to 50% or less.**

**Sensitivity To Detect Infected People**

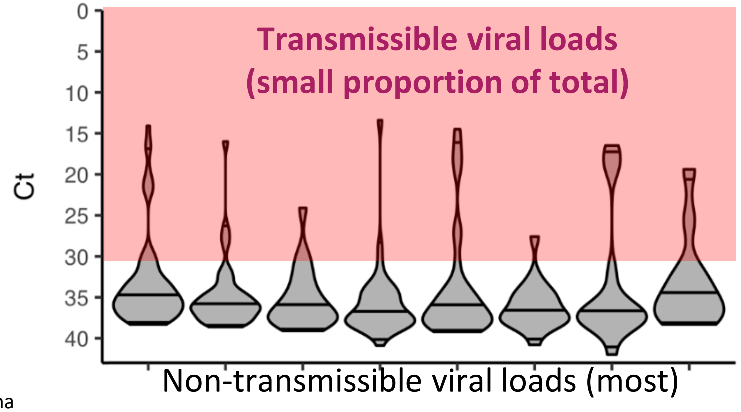
| | 0 days | 1 days | 2 days | 3 days | 4 days | 5 days | |
|---|---|---|---|---|---|---|---|
| **Detected** | 7 | 17 | 21 | 22 | 22 | 24 | |
| **Not Yet Detected** | 13 | 3 | 0 | 0 | 0 | 0 | |
| **Sensitivity** | **37%** | **85%** | **100%** | **100%** | **100%** | **100%** | Correct analysis |
| **Recovered (Ag- but still PCR+)** | **22** | **22** | **22** | **22** | **22** | **22** | |
| **Sensitivity including recovered** | **17%** | **40%** | **48%** | **50%** | **50%** | **52%** | Wrong Analysis |

# WHY DON'T WE ALL HAVE RAPID TESTS TODAY?

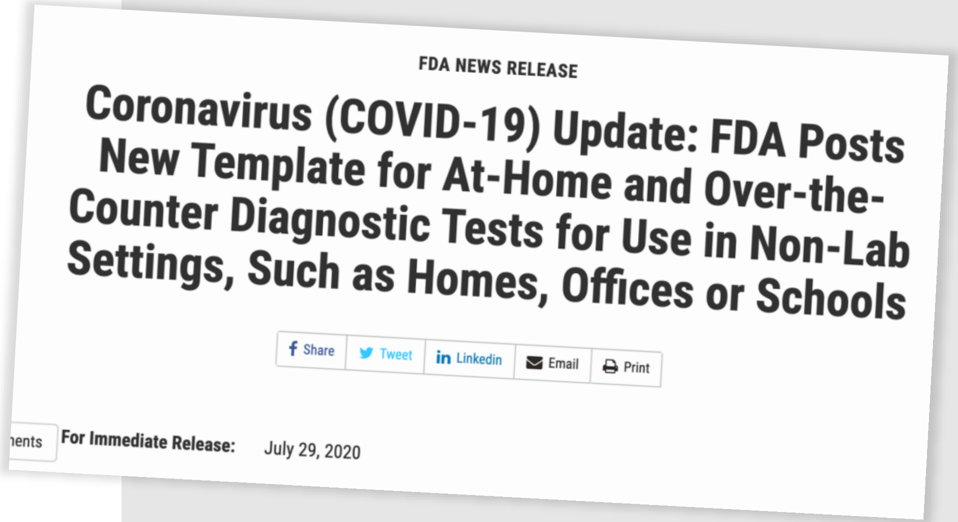



Mina et al, NEJM 2020

## DAILY REPRESENTATIVE SAMPLES OF VIRAL LOADS IN 8 WEEKS IN JUNE/JULY IN MASSACHUSETTS




Graphic: Michael Mina




- **FDA Requires 80% - 90% sensitivity against a "representative sample set" <u>against PCR detection of RNA.</u>**
- For **daily tests**, the <u>**targets for sensitivity should**</u> <u>be</u> <u>**transmissible viral loads**</u>
- **These <u>tests work extremely well to detect infectious virus loads</u>**
- **But MOST PCR positives in asymptomatic "representative sample" are LOW viral loads taken long after transmission ends**

# Alignment with the Biden Strategy: Excerpts from the America Rescue Plan

- *Ensure all Americans have **access to regular, reliable, and free testing**.*
- *Invest in next-generation testing, including **at home tests and instant tests**, so we can scale up our testing capacity by orders of magnitude.*
- *Stand up a Pandemic Testing Board like Roosevelt's War Production Board. It's how we produced tanks, planes, uniforms, and supplies in record time, and it's how we will **produce and distribute tens of millions of tests.***
- *Allocate $50 billion to testing, **including the manufacturing of rapid antigen tests.***

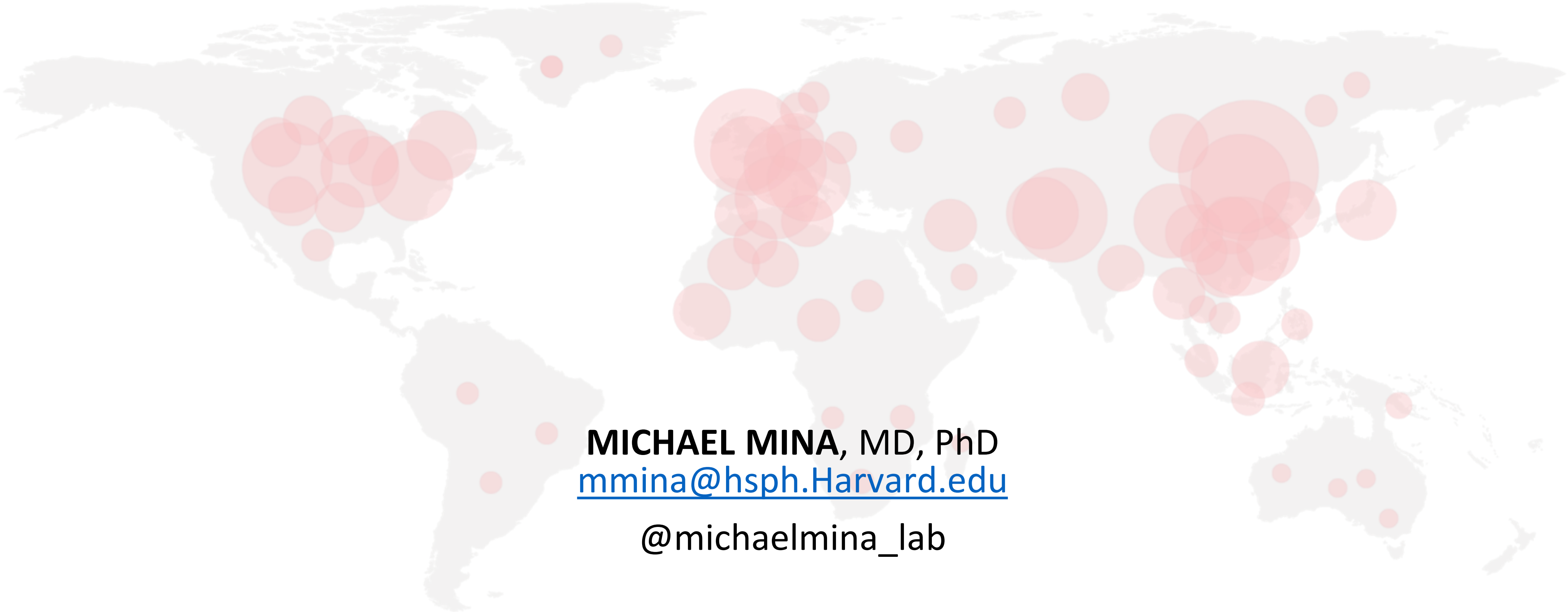
# Obstacles to Implementation

- Regulation (FDA must change its public health approval path)
- Supply (how can we boost supply to 20-30 million tests/day?)
- Distribution (how do we get tests in homes? Schools? Workplaces?)
- Public education (how to administer tests, social behavior risks)
- Public health reporting (can it be voluntary? How do we report?)
- Others

# How the States Can Access Rapid Antigen Testing

- States need to pressure the federal government (especially the FDA) to make inexpensive, OTC/no prescription rapid antigen tests more widely available for asymptomatic population screening
- We need Congress and the Administration to hear from the States that this type of public health screening program can be implemented
- Contact us for more opportunities to collaborate or for assistance with pilot program design.

# REOPENING CITIES, STATES AND SCHOOLS SAFELY
# **VIA FREQUENT RAPID ANTIGEN TESTING**




**MICHAEL MINA**, MD, PhD
mmina@hsph.Harvard.edu

@michaelmina_lab

Center *for* Communicable Disease Dynamics

HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH

HARVARD MEDICAL SCHOOL

BWH Brigham & Women's Hospital